

Advanced (Habanero)

- Q1.** What is the purpose of a two-way frequency table?
- (A) To display one categorical variable
 - (B) To compare two numerical variables
 - (C) To examine the relationship between two categorical variables
 - (D) To calculate the mean of a dataset
 - (E) To create a scatter plot
- Q2.** A survey of 25 students finds that 15 students like pizza and 18 students like ice cream. If 8 students like both, how many students like at least one of them?
- (A) 20
 - (B) 22
 - (C) 25
 - (D) 30
 - (E) 32
- Q3.** In a two-way table, what does a relative frequency of 0.4 mean?
- (A) 40% of the total frequency
 - (B) 4% of the total frequency
 - (C) The frequency is 4
 - (D) The frequency is 0.4
 - (E) None of the above
- Q4.** Which of the following is an example of a two-way frequency table?
- (A) A table showing the number of students in each grade level
 - (B) A table showing the favorite sports of students by gender
 - (C) A graph showing the temperature over time
 - (D) A table showing the average height of students in each class
- (E) A scatter plot of exam scores vs hours studied
- Q5.** If $P(A \cap B) = 0.2$ and $P(A) = 0.5$, what is $P(B|A)$?
- (A) 0.2
 - (B) 0.4
 - (C) 0.5
 - (D) 0.6
 - (E) 0.8
- Q6.** A two-way table shows the relationship between favorite color and eye color. If 60% of the people with blue eyes prefer the color blue, what is the relative frequency of people with blue eyes who prefer blue?
- (A) 0.6
 - (B) 0.4
 - (C) 0.3
 - (D) 0.2
 - (E) 0.1
- Q7.** In a two-way frequency table, the marginal frequency is the
- (A) Frequency of each category
 - (B) Total frequency of each row
 - (C) Total frequency of each column
 - (D) Total frequency of the table
 - (E) Frequency of the intersection of two categories
- Q8.** A survey finds that 30% of the respondents like product A and 20% like product B. If 10% like both, what percentage of respondents like at least one of the products?
- (A) 30%
 - (B) 40%
 - (C) 50%
 - (D) 40%
 - (E) 50%

Open Ended Questions (FRQ)

- Q9.** Create a two-way frequency table to display the relationship between the favorite sport and gender of 50 students. Calculate the relative frequencies and interpret the results.
- Q10.** A company wants to know the relationship between the type of music customers listen to and their age group. Design a two-way frequency table to display this information and explain how to calculate the relative frequencies.

Chef's Tips

- Make sure to provide clear instructions for each question
- Allow students to use calculators for calculations
- Emphasize the importance of labeling axes and titles in graphs